

Program : Diploma in Instrumentation Engineering	
Course Code : 3083	Course Title: Analog Circuits for Instrumentation
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To understand the characteristics of Bipolar junction transistors
- To get the knowledge of the amplifier circuits using transistors
- To get the knowledge of important waveform generator circuits
- To attain knowledge on Operational Amplifiers and its characteristics
- To design simple linear circuits using Operational Amplifiers
- To describe Nonlinear circuits using Operational Amplifiers

Course Prerequisites:

Topic	Course Code	Course name	Semester
Fundamentals of electronics		Applied physics II	2
Fundamentals of Digital		Fundamentals of electrical and electronics	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe the working of transistor amplifier circuits	15	Understanding
CO2	Describe the operation of waveform generator circuits	15	Understanding
CO3	Design linear circuits using operational amplifier	14	Applying
CO4	Explain Non linear circuits using operational amplifier	14	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2						
CO2	2						
CO3	3	2					
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the operations of transistor amplifier circuit		
M1.01	Describe the operation of Bipolar Junction transistor	2	Understanding
M1.02	Compare different transistor configurations	2	Understanding
M1.03	Describe the method of potential divider transistor biasing	2	Understanding
M1.04	Illustrate the working of CE amplifier	2	Understanding
M1.05	Illustrate the Input and Output characteristics of CE amplifier	2	Understanding
M1.06	Describe the frequency response and bandwidth of transistor amplifier	2	Understanding
M1.07	Describe the working of Emitter follower	3	Understanding
Contents: PNP and NPN bipolar junction transistor-Common base, Common Emitter, Common collector configurations –voltage gain, current gain, power gain and I/O impedance of CE configuration –single stage C.E amplifier circuit with potential divider biasing, Input characteristics and output characteristics-AC and DC load line, - frequency response and bandwidth of RC Coupled amplifiers-Working and applications of Common Collector Amplifier/Emitter follower			

CO2	Describe the operations of signal generator Circuits		
M2.01	Explain positive and negative feedback in amplifiers	3	Understanding
M2.02	Describe Barkhausen criteria for oscillators	2	Understanding
M2.03	Explain the working of different oscillator circuits	7	Understanding
M2.04	Describe the working of astablemultivibrator using transistors	3	Understanding
	Series Test – I	1	
Contents: Feedback amplifier- positive and negative feedback- Gain of the feedback amplifier–Barkhausen criteria- Oscillators-Classification -: RC phase shift oscillator—Wien bridge oscillators, LC oscillators- Hartley oscillator- Colpitt’soscillator-crystal oscillator-AstableMultivibrator - expression for frequency of Oscillation			
CO3	Design linear circuits using operational amplifier		
M3.01	Describe the block diagram and parameters of operational amplifier	3	Understanding
M3.02	Distinguish ideal op amp characteristics and practical op-amp characteristics	1	Understanding
M3.03	Describe important op amp parameters(Voltage gain, Gain bandwidth product, Input impedance, output impedance, Input bias current, Input offset voltage, Input offset current)current, slew rate, CMRR)	2	Understanding
M3.04	Illustrate linear applications of op amp	5	Applying
M3.05	Develop linear op amp circuits for simple instrumentation applications (Amplification, Integration, Differentiation)	3	Applying
Contents: Operational Amplifier : Op-amp symbol — block diagram- Ideal Op-amp properties - Properties of practical op-amp–Op-amp parameters(Voltage gain, Gain bandwidth product,Input impedance, output impedance, Input bias current, Input offset voltage, Input offset current, slew rate, CMRR) Linear Applications of operational amplifier-Inverting amplifier – Non-inverting amplifier – Voltage follower-Simple problems to design amplifiers-.Summing amplifier - difference amplifier - - adder - sub tractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter – Integrator – Differentiator – simple problems on designing inverting and non-inverting amplifiers for given gain.			

CO4	Explain Nonlinear circuits using operational amplifier		
M4.01	Illustrate the circuit of precision rectifier using op-amp	1	Understanding
M4.02	Describe different comparator circuits	5	Understanding
M4.03	Illustrate clipper and clamper circuits	4	Understanding
M4.04	Explain sample and hold circuit	2	Understanding
M4.05	Illustrate log and antilog amplifier	2	Understanding
	Series Test - II	1	
Contents: Non Linear Applications of Op –amps-Precision rectifiers – Peak detector – Zero crossing detector Comparators – Regenerative comparator (Schmitt trigger) -Clipper and Clamper – Sample and Hold circuit-Log and antilog amplifier.			

Text / Reference:

T/R	Book Title/Author
T1	J.B. Gupta, Electronic circuits and devices, S. K. Kataria and sons.
R1	Roy Choudhary, Linear integrated circuits, New Age publications.
R2	Floyd , Electronic devices , Pearson education.
R3	Millman and Halkias, Electronic devices and circuits, TataMcGraw-Hill
R4	Ramakant Gayakward , Op-amps and Linear Integrated Circuits , Pearson education

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in › content › storage2 › courses
2	https://www.electronicshub.org › ic-741-op-amp-basics
3	https://nptel.ac.in/content/storage2/courses/downloads/108108125/noc19_ee32_Assignment1.pdf
4	https://www.tutorialspoint.com/computer_logical_organization/digital_counters.htm
5	https://www.allaboutcircuits.com/worksheets/counters/